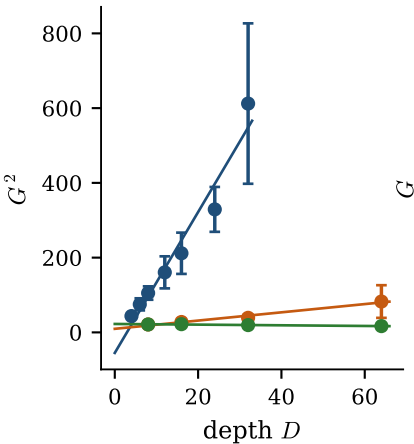
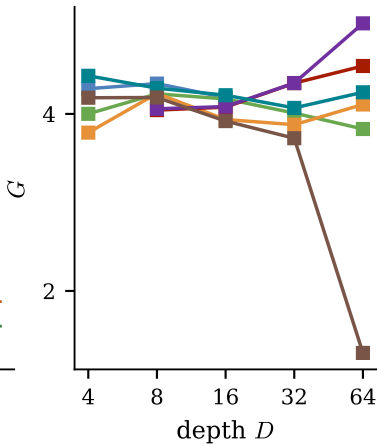
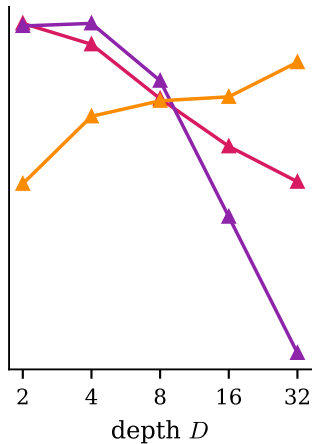


(a) MLPs: G^2 grows with D 

(b) CNNs (CIFAR-10)



(c) ViTs (CIFAR-10)



MLP plain (slope/ $s^2 = 18.8$)

MLP res. $s=1/\sqrt{8}$ (slope/ $s^2 = 8.8$)

MLP res. $s=1/\sqrt{D}$

CNN plain

CNN res. $s=1/\sqrt{D}$

CNN res. $s=1/\sqrt{8}$

CNN res. $s=1$

CNN plain (Dirac)

CNN pre-norm res.

CNN post-norm plain

ViT pre-LN

ViT pre-LN, $s=1/\sqrt{D}$

ViT post-LN